Cardiovascular Diabetology

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Improved prediction and risk stratification of major adverse cardiovascular events using an explainable machine learning approach combining plasma biomarkers and traditional risk factors

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Abstract

Background Cardiovascular diseases (CVD) remain the leading cause of morbidity and mortality globally. Traditional risk models, primarily based on established risk factors, often lack the precision needed to accurately predict new-onset major adverse cardiovascular events (MACE). This study aimed to improve prediction and risk stratification by integrating traditional risk factors with biochemical and metabolomic biomarkers.

Methods We analyzed data from 229,352 participants in the UK Biobank (median age 58.0 years; 45.4% male) who were free of baseline MACE. Biomarker selection was conducted using area under the curve (AUC), minimal joint mutual information maximization (JMIM), and correlation analyses, while Cox proportional hazards models were employed to evaluate the predictive performance of combined traditional risk factors and biomarkers. Optimal binary thresholds were determined utilizing CatBoost and SHAP, leading to the calculation of a Biomarker Risk Score (BRS) for each participant. Multivariable Cox models were conducted to assess the associations of each concerned biomarker and BRS with new-onset endpoints.

Results The combination of PANEL + All Biochemistry + Cor0.95 of Nonov Met predictors demonstrated significantly improved discriminative performance compared to traditional models, such as Age + Sex and ASCVD, across all endpoints. Although the prediction for hemorrhagic stroke was suboptimal (C-index = 0.699), C-index values for other outcomes surpassed 0.75, with the highest value (0.822) recorded for CVD-related mortality. Key predictors of new-onset MACE included cystatin C, HbA1c, GlycA, and GGT, while IGF-1 and DHA exhibited potential protective effects. The BRS stratified individuals into low-, intermediate-, and high-risk groups, with the strongest effect observed for CVD death, where the high-risk group had a relative risk of 2.76 (95% CI 2.48–3.07) compared to the low-risk group.

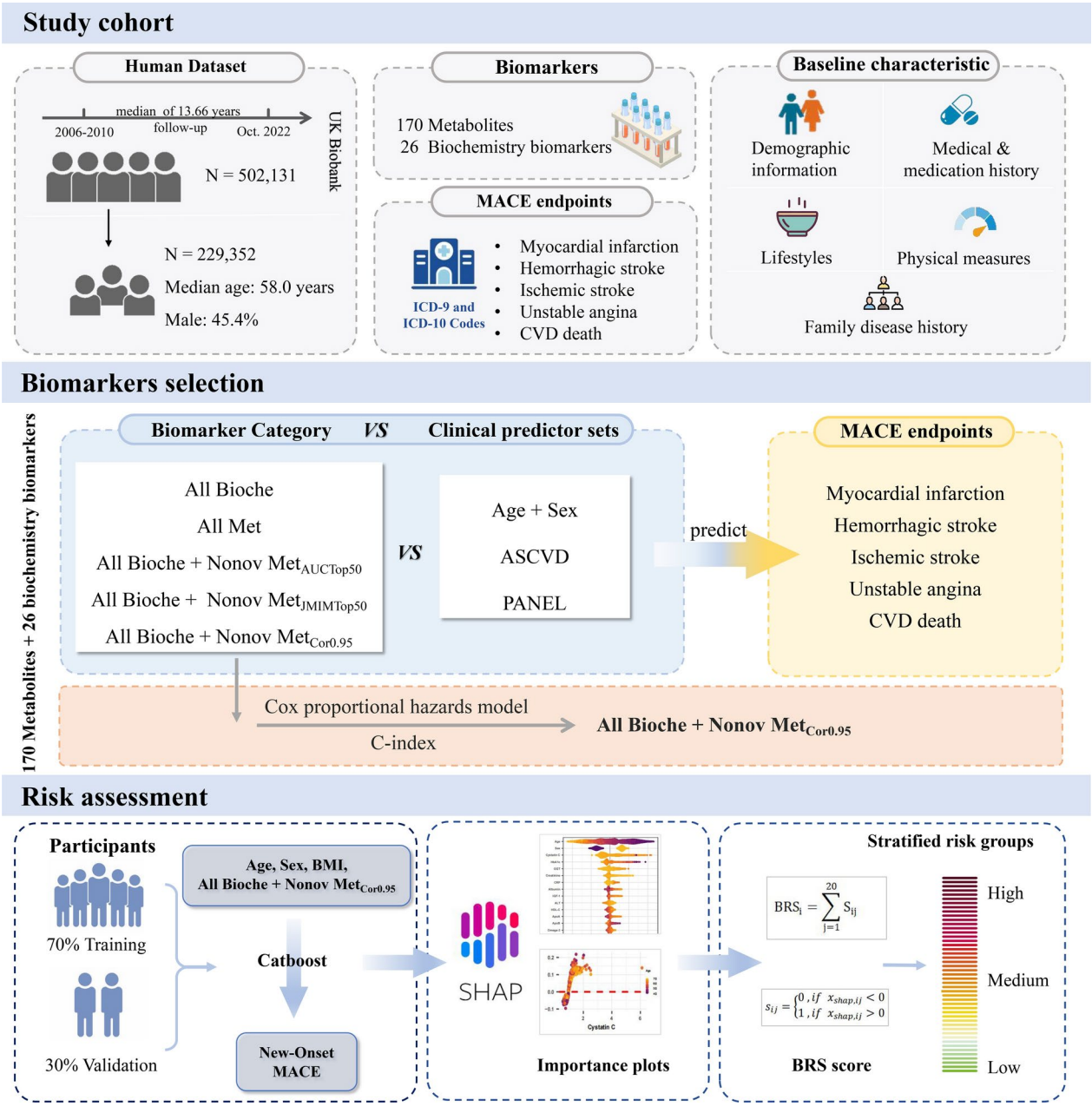
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Conclusion Integrating traditional risk factors and biomarkers improves prediction and risk stratification of new-onset MACE. The BRS shows promise as a tool for identifying high-risk individuals, with the potential to support personalized CVD prevention and management strategies.

Graphical abstract



Keywords Cardiovascular research, Major adverse cardiovascular events, Biomarker risk score, Risk stratification, Binary threshold

Introduction

Cardiovascular diseases (CVD) remain the leading cause of mortality and disability worldwide, posing a significant burden on public health systems and societies at large [1, 2]. Effective prevention and management of CVD are contingent upon accurate prediction and risk stratification [3]. Traditional risk prediction models, such as the ESC SCORE2 and AHA-ASCVD, have provided essential

tools for assessing CVD risk based on established clinical and demographic factors, including age, sex, blood pressure, cholesterol levels, smoking status, and diabetes [4, 5]. However, despite their widespread adoption in clinical practice, these models often neglect emerging biomarkers that play pivotal roles in CVD pathogenesis. Key among these biomarkers are glycated hemoglobin (HbA1c), glycoprotein acetyls (GlycA), cystatin C and omega-3, which have been independently associated with CVD outcomes and related mortality [6–13]. The failure to routinely incorporate these biomarkers into risk prediction models underscores a critical gap in our understanding of CVD risk and its modifiable components.

Recent advances in high-throughput technologies, particularly nuclear magnetic resonance (NMR) spectroscopy, have transformed the field of metabolomics, enabling the comprehensive and standardized quantification of a diverse array of small molecules circulating in the blood [3, 12]. This capability allows for an in-depth exploration of the complex metabolic pathways underlying disease processes. The ability to profile numerous metabolites simultaneously presents a unique opportunity to identify novel biomarkers and enhance our understanding of their roles in disease etiology. In the context of CVD, these technological advancements have broadened our perspective beyond traditional risk factors, uncovering a more extensive spectrum of metabolic signatures associated with cardiovascular health [3, 12, 14, 15]. Nevertheless, the systematic integration of clinical, metabolic, and traditional risk factors to enhance CVD prediction remains an underexplored domain. Furthermore, the interpretability and clinical relevance of these biomarkers in relation to specific CVD outcomes, such as major adverse cardiovascular events (MACE), have yet to be fully elucidated.

This study seeks to address this gap by integrating clinical biochemistry markers, NMR-based metabolomics data, and traditional CVD risk factors to enhance the prediction of incident MACE. Utilizing data from the UK Biobank—a large, population-based cohort—we assess the predictive performance of these combined biomarkers through Cox proportional hazards models and machine learning techniques, specifically SHapley Additive exPlanations (SHAP), to identify the most significant predictors and their optimal thresholds [16–18]. Our approach not only advances our understanding of the individual and collective roles of these biomarkers but also facilitates the development of a biomarker risk index (BRI) and biomarker risk score (BRS), which could significantly refine risk stratification in clinical practice. Ultimately, this study offers novel insights into the integration of innovative biomarkers into CVD risk prediction models, paving the way for the early identification and intervention of high-risk individuals.

Methods

Study designs and populations

The UK Biobank is an ongoing population-based prospective cohort study that features extensive phenotypic data and biomarkers measure. Between 2006 and 2010, more than 500,000 participants aged 40–69 years from the general population were recruited at 22 assessment centers across the United Kingdom. Sociodemographic information, lifestyle and environmental factors, medications information and family history were collected through touchscreen questionnaire or verbal interview at recruitment. Pre-existing health and medical conditions were obtained from the baseline questionnaire and interview, as well as hospital records and death registries. Anthropometric measurements and blood sample collection and laboratory measures were taken at recruitment. The protocol for handling and storing blood samples has been detailed in prior publications [3, 6, 18, 19]. For the purpose of our analyses, we excluded individuals with any type of MACE or entirely missing biochemistry marker data at baseline from those randomly selected for NMR-based metabolomic profiling, resulting in an analytic cohort of 229,352 participants.

Conventional predictor and biomarker set

The overview of the study workflow, as well as clinical predictor sets and biomarker sets were presented in Fig. 1. We investigated three sets of conventional clinical predictor variables: Age + Sex, ASCVD and PANEL. Cardiovascular predictors set (ASCVD) were selected according to the cardiovascular risk scores for primary prevention recommended by the ESC and AHA, including the ESC SCORE2 and the AHA-ASCVD score [4, 5]. The PANEL predictor set included a comprehensive range of demographics, lifestyle, physical measurements, medical history, medications and the CVD family history.

A list of all clinical predictors applied in this study is presented in Supplementary Table 1 and a list of all metabolomic predictors in Supplementary Table 2. A total of 30 conventional clinical biochemistry markers were measured from blood samples collected at recruitment. After excluding variables with over 20% missing values, substantial sex differences, or substantial variations based on fasting duration, 26 biochemical markers were included in the analysis. Furthermore, a random subset of 274,236 participants was characterized using a high-throughput NMR metabolomics platform developed by Nightingale Health Ltd. The NMR assay covers 170 original biomarkers spanning multiple metabolic pathways including amino acids, fatty acids, glycolysis metabolites, ketone bodies, inflammation markers as well as cholesterol subtypes and lipoprotein lipids in 14 subclasses [3, 12]. For consistency, the same exclusion criteria applied to the conventional clinical biochemistry

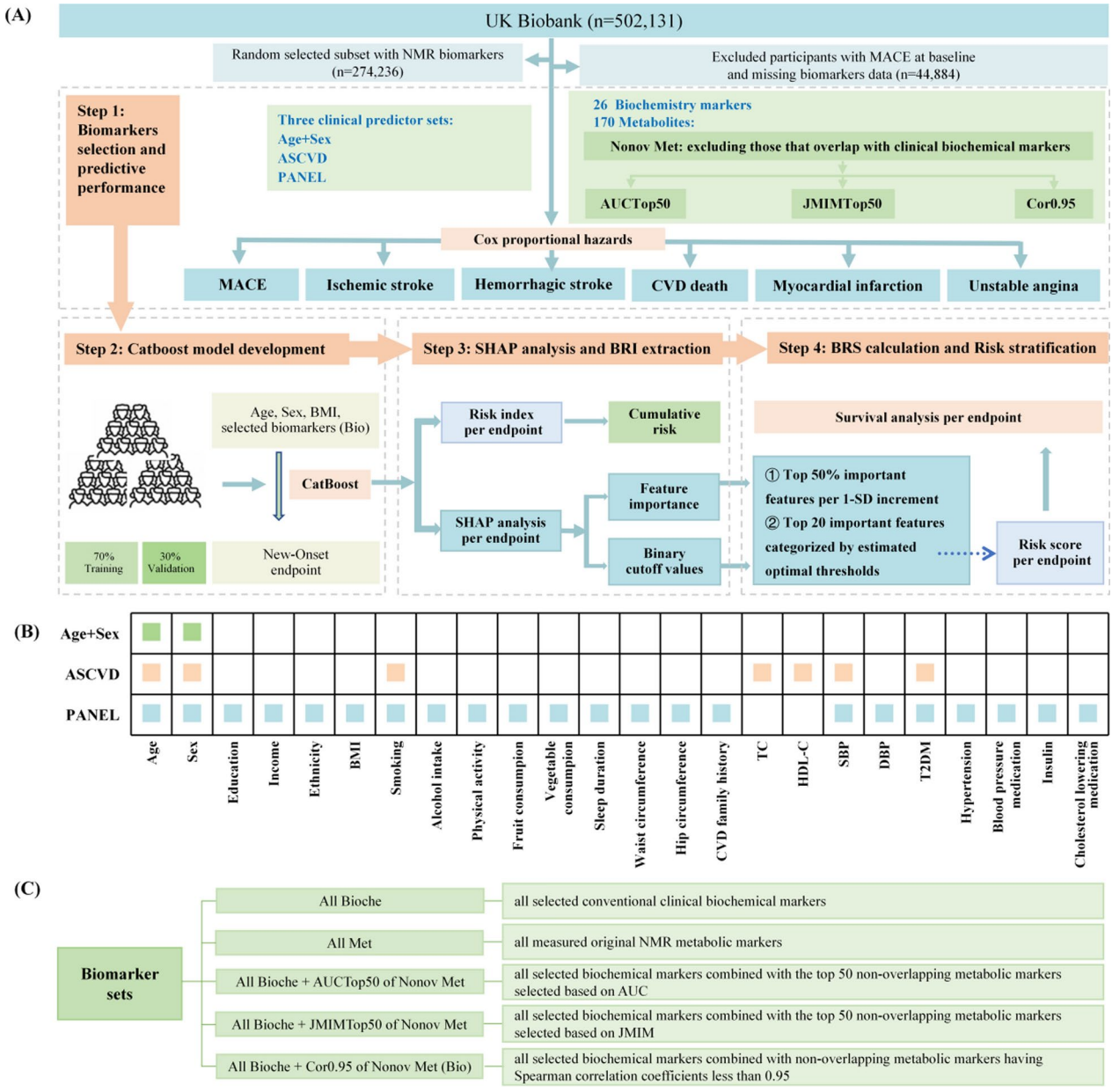


Fig. 1 Study overview. ASCVD: Age, Sex, Smoking, T2DM, SBP, TC, HDL-C; PANEL: Age, Sex, Education, Income, Ethnicity, BMI, Smoking status, Alcohol intake, Physical activity, Fruit consumption, Vegetable consumption, Sleep duration, Waist circumference, Hip circumference, CVD family history, SBP, DBP, T2DM, Hypertension, Blood pressure medication, Insulin, Cholesterol lowering medication. *NMR* nuclear magnetic resonance, *MACE* major adverse cardiovascular events, *BMI* body mass index, *SHAP* SHapley Additive exPlanations, *BRI* biomarker risk index, *BRS* biomarker risk score, *SD* standard deviation, *CVD* cardiovascular disease, *TC* total cholesterol, *HDL-C* HDL cholesterol, *SBP* systolic blood pressure, *DBP* diastolic blood pressure, *T2DM* type 2 diabetes mellitus, *Bioche* biochemical markers, *Met* metabolic markers, *Nonov Met* high-throughput nuclear magnetic resonance (NMR) metabolic markers, excluding those that overlap with clinical biochemical markers, *AUC* receiver operating characteristic curve, *JMIM* minimal joint mutual information maximization, *Cor0.95* Spearman correlation coefficients less than 0.95, *Bio* a total of 26 biochemical markers combined with Cor0.95 of Nonov Met

markers were also applied to the NMR metabolomic biomarkers. Specifically, missing values for any metabolite did not exceed 20%. For the clinical biochemistry markers, only testosterone was excluded due to substantial sex differences (Supplementary Table 3), which led to the information provided by this marker in the SHAP dependence plot being primarily attributed to sex differences (Supplementary Fig. 1). No similar issues were observed with the NMR metabolomic biomarkers.

MACE outcome definition

The new-onset MACE was defined as a composite of the following events: incident myocardial infarction, incident stroke classified as either ischemic or hemorrhagic, death

from CVD, and the occurrence of unstable angina, representing a range of serious cardiovascular outcomes that may significantly impact patient prognosis [9, 20]. We investigated the composite endpoint defined as MACE, along with each of its components, derived from hospital episode statistics and national death registries information. Incident and mortality outcomes were determined based on the first documented occurrence of relevant events, identified using the International Classification of Diseases, 10th revision (ICD-10) or ICD-9 codes [13, 21]. The follow-up of hospitalizations ended on October 31, 2022 in England, August 31, 2022 in Scotland, and May 31, 2022 in Wales. The follow-up for the death registry ended on November 30, 2022. Follow-up visits lasted from the initial attendance at assessment center to the earliest date of relevant outcome diagnosis, death, or the latest hospital inpatient data, whichever occurred first.

Statistical analysis

Data imputation and baseline characteristics summary

For the potential conventional risk factor, we assumed that the missing data occurred at random and conducted multiple imputations using chained equations with random forests [22, 23]. This approach generated five datasets, from which one was randomly selected for subsequent analysis. Missing values in any covariate were less than 20%. To minimize the impact of outliers for the original metabolites, plasma biomarker values exceeding four interquartile ranges from the median were replaced with their corresponding boundary values [12]. Missing values for any metabolite were below 5% and were imputed using a random forest algorithm. The baseline characteristics of the overall population and specific datasets (incident MACE, stroke, death from CVD, myocardial infarction, unstable angina) were presented as median (interquartile range [IQR]) for continuous variables, and number (percentage [%]) for categorical variables.

Biomarkers selection and model predictive performance

To preliminarily identify biomarkers for enhancing prediction and risk stratification of new-onset MACE, we considered several sets of biomarkers, each incorporating different combinations of clinical biochemical and metabolic markers. These sets included: (1) all selected conventional clinical biochemical markers (All Bioche), (2) all measured original NMR metabolic markers (All Met), (3) all selected biochemical markers combined with the top 50 non-overlapping metabolic markers selected based on the area under the receiver operating characteristic curve (AUC) (All Bioche + AUCTop50 of Nonov Met), (4) all selected biochemical markers combined with the top 50 non-overlapping metabolic markers selected based on minimal joint mutual information maximization (JMIM) values (All Bioche + JMIMTop50 of Nonov Met), and (5) all selected biochemical markers combined with non-overlapping metabolic markers having Spearman correlation coefficients less than 0.95 (All Bioche + Cor0.95 of Nonov Met) [24]. The naming of the biomarker sets and their contents is further illustrated in Fig. 1C.

These sets were incorporated into Cox proportional hazards model, and Harrell's C-index along with 95% confidence interval (CI) was employed to assess model's discrimination performance. The 95% CI was calculated using the bootstrap method, with resampling performed 1000 times [25]. Non-overlapping metabolic markers were selected using the Filter method from the *mlr3* package, which serves as a preprocessing step prior to model training. For the AUCTop50 of Nonov Met set, we applied the flt("auc") method to separately calculate the AUC for each non-overlapping metabolite in predicting the target variable. The top 50 metabolites with the highest AUC values were then selected. Similarly, for the JMIMTop50 of Nonov Met, we used the flt("jmim") method to separately compute JMIM score for each feature, which indicates the degree of information shared between independent variable and the target variable. The top 50 features with the highest JMIM score were selected. In the Cor0.95 of Nonov Met subset, we performed a correlation-based variable selection approach to identify and eliminate highly correlated predictors. We computed the Spearman correlation matrix among all non-overlapping metabolites and applied a threshold of 0.95 to identify variable pairs with high correlation. Variables with correlation coefficients of 0.95 or higher were considered redundant. To improve computational efficiency, we used an exact method for correlation matrices with fewer than 100 variables and an approximate method for larger matrices. This strategy effectively reduced multicollinearity, ensuring the robustness of our subsequent analyses. Ultimately, 48 metabolites with Spearman correlation coefficients less than 0.95 were retained. Taking into account both performance and complexity of established model, we selected combined "All Bioche + Cor0.95 of Nonov Met" set as biomarkers for the risk stratification of new-onset MACE and each of its individual components. Subsequently, we modeled disease risk for each endpoint using Cox proportional hazards models based on the previously defined conventional clinical predictor sets: Age + Sex, ASCVD, and PANEL. For all sets, the performance of the Cox proportional hazards models was benchmarked against those based on the combinations of these conventional sets with the selected biomarkers sets, which including 26 biochemistry and 48 metabolic markers, making us to assess whether the selected biomarkers could improve the predictive performance of the models [3].

Feature importance and BRI extraction

The selected biomarkers, along with sex, age, and body mass index (BMI), were incorporated as features to develop CatBoost classifiers for predicting incident MACE and each of its components. While the relationship between BMI and vascular health can be complex and potentially non-linear, given the well-established association between BMI and cardiovascular outcomes [21, 26–30], as well as its role as a key indicator of body composition, and its correlation with various metabolic factors [31, 32], we believe BMI should be considered a critical variable in cardiovascular risk prediction. CatBoost is an unbiased boosting algorithm with support for categorical features. It processes data using ordered boosting and permutation-based statistics, which helps minimize prediction shift, reduces overfitting, and improves model generalization. Additionally, CatBoost's optimized GPU implementation enables faster training on large datasets, making it scalable for large-scale tasks. Empirical results have demonstrated that CatBoost consistently outperforms other gradient boosting models, such as XGBoost and LightGBM, both in terms of accuracy and efficiency. These advantages, combined with its ability to provide interpretable results through SHAP values, make CatBoost a reliable and efficient choice for practical machine learning applications, especially in fields requiring high transparency and accuracy [33–35].

For each endpoint, the data was randomly split into training (70%) and validation (30%) sets. Hyperparameter tuning and model fitting were conducted on the training set, while the validation set was used to evaluate model performance and determine optimal hyperparameters. The final model was then fitted on the entire dataset using the identified the optimal hyperparameters, and the SHAP value [17, 33, 36] were calculated to interpret the output of CatBoost models [33–35]. SHAP values fairly distribute the "contribution" of each feature to the model's prediction, quantifying how much each feature increases or decreases a specific prediction compared to the average across the dataset. Feature importance was visualized using the SHAP summary plot, which aggregates SHAP values to identify the most influential predictors in the model. The BRI is derived from the probabilities predicted by the CatBoost classifier for each endpoint. Specifically, the CatBoost classifier outputs the risk probabilities for the target variable, where each individual's risk is expressed as a probability ranging between 0 and 1. These probabilities represent the likelihood that an individual will experience the endpoint of interest. For each endpoint, we categorized individuals into five risk groups based on the quintiles of the BRI and plotted the cumulative risk for individuals in the first, third, and fifth quintile groups.

Notably, the CatBoost model we employed utilizes gradient boosting decision trees, which inherently capture complex relationships between variables through the splitting mechanism. This is particularly useful for handing high-dimensional or nonlinear data. As a result, even with a nonlinear relationship between continuous variables and the target variable, CatBoost can automatically learn these patterns without the need for additional feature engineering [17]. To explore potential non-linear relationships between BMI and new-onset MACE, as well as its components, we conducted sensitivity analyses by fitting restricted cubic splines to a multivariable-adjusted Cox proportional hazards model (RCS-Cox). We evaluated the Akaike Information Criterion (AIC) for models with 3–7 knots and selected the model with 3 knots (at the 10th, 50th, and 90th percentiles) based on the principle of minimizing AIC.

SHAP-based feature interpretation, BRS and risk assessment

We first identified several meaningful variables that were consistently ranked in the top 50 of importance for predicting different types of MACE outcomes using CatBoost classifiers and SHAP analysis. The multivariable Cox proportional hazards model, adjusted for age, sex, and BMI, was then applied to estimate the risk of new-onset MACE and its components for each one standard deviation (SD) change in these variables. Next, SHAP dependence plots were generated to visualize the importance of features, along with their approximate optimal binary thresholds and the direction associated with MACE and each of its components. The X-axis represents the original values of each given feature, while the Y-axis shows the corresponding SHAP value. A SHAP value greater than zero indicates that the feature contributes to a higher probability prediction (e.g., predicting '1'); while the SHAP value less than zero drives the prediction toward a lower probability (e.g., predicting'0'). That is to say a SHAP value above zero suggests that the feature may increase the risk of new-onset MACE. Based on whether the SHAP value is greater than or equal to zero, we categorize the feature as a binary variable. Each feature was then binarized based on its estimated optimal threshold. We further applied the Cox proportional hazards model, adjusted for sex, age, and BMI, to evaluate the association between the top 20 features (categorized as greater than or equal to, or less than, their respective optimal thresholds) and the risk of incident MACE and its components, compared to values below the threshold.

Next, we calculated the BRS for MACE and its components for each individual, using the top 20 features and their respective optimal binary thresholds identified through SHAP analysis [16, 37]. The formula for BRS is as follows:

$$\mathrm{BRS_i} = \sum_{\mathrm{j=1}}^{20} \mathrm{S_{ij}}$$

where $\mathrm{BRS_i}$ represents the BRS for individual i. $\mathrm{S_{ij}} = \begin{cases} 0, & \text{if } X_{shap,ij} < 0 \\ 1, & \text{if } X_{shap,ij} \geq 0 \end{cases}$, which denotes the binary risk score assigned to the *j*-th feature for the *i*-th individual. The binary risk score for each feature is assigned based on the association with MACE or its components, as identified through CatBoost classifiers and SHAP analysis. The top 20 features identified are used for calculating the BRS. The $\mathrm{X_{shap,ij}}$ is the estimated SHAP value for the *j*-th feature of the *i*-th individual.

For a participant, if a feature exhibits a positive association or a trend toward a positive relationship with the target variable (MACE and each of its components), a score of 1 is assigned when the feature's original value is greater than or equal to the binary cutoff value defined in the SHAP dependent plot (in this case, the feature's estimated SHAP value is located in the upper-right portion of the SHAP dependence plot, greater than or equal to 0); otherwise, the score is set to 0. Conversely, if a feature demonstrates a negative association or a trend toward a negative relationship, a score of 1 is assigned when the original feature value is less than the cutoff value (in this case, the feature's estimated SHAP value is located in the upper-left portion of the SHAP dependence plot, greater than or equal to 0); the score is 0 otherwise.

For example, since age is positively associated with the risk of incident MACE, individuals aged 60 years or older (if 60 is the binary cutoff value for age) would be assigned a score of 1, while those younger than 60 years would receive a score of 0. In contrast, if insulin-like growth factor 1 (IGF-1) exhibits a negative association with incident MACE, individuals with IGF-1 levels greater than or equal to the cutoff value would be assigned a score of 0, while those with levels below the cutoff value would receive a score of 1. To illustrate this, suppose an individual has the following binary scores for the top 20 features, feature1 = 1, feature2 = 0, feature3 = 1, feature4 = 1, ⋯feature20 = 0, the BRS for this individual would be:

$$\begin{aligned}\mathrm{BRS} = {} & 1\,(\text{Feature 1}) + 0\,(\text{Feature 2}) + 1\,(\text{Feature 3}) \\ & + 1\,(\text{Feature 4}) + \cdots + 0\,(\text{Feature 20})\,.\end{aligned}$$

Study participants were then classified into low-, moderate-, and high-risk groups according to the tertiles of the BRS for further analysis. We calculated both the incidence proportion and incidence density (ID), which represent the absolute risk [38–40], for MACE and its components across low-, medium-, and high-risk groups based on the BRS. The number of new-onset cases and the incidence proportion, also referred to as cumulative incidence, were presented as No. of cases (%), calculated using the following formula:

$$\begin{aligned}&Incidence\ proportion \\ &= \frac{Number\ of\ new\ cases\ of\ diseasein\ a\ specific\ period}{Total\ population\ at\ risk}\end{aligned}$$

The ID, also referred to as incidence rate, was calculated using the formula:

$$\begin{aligned}&Incidence\ density \\ &= \frac{Number\ of\ new\ cases\ of\ disease\ in\ a\ specific\ period}{Total\ person-time\ of\ observation}\end{aligned}$$

Total person-time was defined as the sum of the time periods of observation each person. A multivariable-adjusted Cox proportional hazards model was used to estimate hazard ratios (HR) and 95% CI for incident MACE and each of its components in the moderate- and high-risk groups, with the low-risk group serving as the reference. The model was adjusted for a comprehensive set of covariates, including demographic factors (age, sex, ethnicity, education, and the average household income), lifestyle factors (smoking status, alcohol consumption frequency, weekly minutes of moderate physical activity, total fruit intake, and total vegetable consumption), and health condition and medication history (systolic blood pressure [SBP], diastolic blood pressure [DBP], antihypertensive medication use, insulin use, cholesterol-lowering medication use, the family history of CVD, and the presence of type 2 diabetes mellitus [T2DM] and hypertension at baseline).

Additionally, SHAP dependence plots enable the visualization of interaction effects between features. For each feature, the plot not only shows its individual contribution but also illustrates how its impact on the prediction changes with the values of other features. The color gradient typically represents the interaction with another key feature, such as sex, revealing how the relationship between a given feature and the endpoint varies depending on the values of other features [17]. Furthermore, stratified analyses were conducted by sex using a multivariable-adjusted Cox proportional hazards model. The interactions between stratification factors and each biomarker of interest with respect to the endpoint were assessed using the likelihood ratio test.

All statistical analyses were performed using R software version 4.3.1 (R Development Core Team, Vienna, Austria). All tests in our study were two-sided. *P*-values were adjusted for multiple comparisons using the Benjamini-Hochberg (BH) method to control the false discovery rate (FDR). The adjusted *P*-values are reported as P-FDR. A P-FDR of < 0.05 was considered statistically significant.

Results

Study population characteristics

The study included 229,352 participants who were free of MACE at baseline, with a median age of 58.0 years (interquartile range [IQR]: 50.0–63.0). Of these participants, 45.4% were male, and 95.1% were of White ancestry. Over a median follow-up period of 13.66 years (IQR: 12.79–14.35), 16,333 participants experienced a confirmed first MACE. Individuals who developed MACE were more likely to have lower educational attainment and income levels, as well as higher plasma concentrations of c-reactive protein (CRP), HbA1c, and triglycerides (TG), in addition to higher waits circumference, hip circumference, body mass index (BMI), systolic blood pressure (SBP), and diastolic blood pressure (DBP). Furthermore, individuals who developed MACE were more likely to have prevalent hypertension or type 2 diabetes mellitus (T2DM) at baseline and to report a family history of cardiovascular disease (Table 1).

Superior discriminative performance of combined biomarkers in cardiovascular risk stratification

Various methods were employed to select biomarkers, with predictive performance evaluated using Harrell's C-index and corresponding 95% CI. When the Cox proportional hazards model was constructed using the feature subset of All Bioche + Cor0.95 of Nonov Met, the model achieved a C-index of 0.698 (95% CI 0.695–0.703), slightly outperforming models that included All Bioche + AUCTop50 of Nonov Met (C-index = 0.692, 95% CI 0.689–0.697) and All Bioche + JMIMTop50 of Nonov Met (C-index = 0.692, 95% CI 0.690–0.698). For models incorporating the PANEL variables, the combination of PANEL + All Bioche + Nonov Met and PANEL + All Bioche + Cor0.95 of Nonov Met yielded C-indices of 0.753 (95% CI 0.751–0.758) and 0.752 (95% CI 0.749–0.756), respectively. Given its superior performance, reduced complexity, and lower multicollinearity, All Bioche + Cor0.95 of Nonov Met was selected as the biomarker subset for subsequent analyses (Supplementary Fig. 2).

Incorporating the selected biomarkers into prediction models resulted in a substantial improvement in discriminative performance for all endpoints. The full model, combining PANEL predictors with the selected biomarkers, demonstrated enhanced predictive capability across all cardiovascular outcomes, outperforming the base models. While the predictive performance for hemorrhagic stroke was suboptimal (C-index = 0.699; 95% CI 0.697–0.719), C-indices for other endpoints exceeded 0.75, with the highest performance observed for predicting CVD-related death (C-index = 0.822, 95% CI 0.816–0.826). These results underscore the value of incorporating biomarkers to improve model accuracy and refine risk stratification for cardiovascular outcomes (Fig. 2).

CatBoost classifier and SHAP analysis reveals key factors influencing the prediction of MACE

To assess the impact of selected biomarkers, along with age, sex, and BMI, on CatBoost model predictions, we employed SHAP importance plots. Age was identified as the most influential factor in predicting MACE. Older participants (represented in dark purple) were more likely to develop MACE (right side), whereas younger participants (depicted in orange-yellow) had a lower likelihood of experiencing MACE and were more likely to remain healthy (left side). Cystatin C ranked third in importance, with higher levels correlating with an increased risk of incident MACE. In contrast, albumin and insulin-like growth factor 1 (IGF-1) ranked eighth and ninth, with higher levels linked to a reduced risk, while lower levels were associated with an increased risk (Fig. 3A).

SHAP dependence plots were utilized to visualize how changes in individual features affected model predictions and to explore the interactions between features. For MACE prediction, individuals aged 60 years or older had higher SHAP values, indicating a greater risk of new-onset MACE compared to those younger than 60 years (Fig. 3B). Similarly, cystatin C levels ≥ 1.0 mg/L (Fig. 3D) or HbA1c concentrations ≥ 39.0 mmol/mol (Fig. 3E) were associated with an increased risk of MACE. Figure 3J illustrates the relationship between IGF-1 levels and SHAP values for MACE prediction. Lower IGF-1 levels (< 18.0 nmol/L) were associated with higher SHAP values, indicating an increased risk of MACE, whereas higher IGF-1 levels were linked to lower SHAP values, demonstrating a reduced risk of MACE. We observed potential interactions between several variables, such as age and cystatin C, sex and cystatin C, as well as creatinine and sex. Stratified analyses suggested that the associations of cystatin C, creatinine, with the risk of incident MACE differed by sex. The likelihood ratio test indicated significant interactions between these variables and sex (Supplementary Table 4). Detailed information on the importance and interpretability of each MACE component is provided in the supplementary materials (Supplementary Fig. 3–7).

Cox proportional hazards model identifies key biomarkers associated with new-onset MACE and its components

In the prediction of new-onset MACE and their individual components, key variables such as age, sex, cystatin C, HbA1c, gamma-glutamyltransferase (GGT) and CRP ranked among the leading 20 most important predictors, while biomarkers such as creatinine, HDL cholesterol (HDL-C), GlycA, urea, alanine aminotransferase (ALP), docosahexaenoic acid (DHA), phospholipids in medium

Table 1 Baseline characteristics of study participants by incident MACE and each of its components

Characteristics	Overall (N = 229,352)	Incident MACE (N = 16,333)	Incident Stroke (N = 5,977)	Incident CVD death (N = 6,229)	Incident Myocardial Infarction (N = 6,093)	Incident Unstable angina (N = 1,880)
Age at recruitment (years)[a]	58.0 [50.0, 63.0]	62.0 [57.0, 66.0]	63.00 [58.0, 66.0]	64.00 [59.0, 67.0]	61.00 [55.0, 65.0]	61.00 [55.0, 65.0]
Sex						
Male	104,088 (45.4)	10,232 (62.6)	3,344 (55.9)	3,996 (64.2)	4,169 (68.4)	1,176 (62.6)
Female	125,264 (54.6)	6,101 (37.4)	2,633 (44.1)	2,233 (35.8)	1,924 (31.6)	704 (37.4)
Ethnicity						
White	218,050 (95.1)	15,611 (95.6)	5,745 (96.1)	5,984 (96.1)	5,804 (95.3)	1,762 (93.7)
Others	11,302 (4.9)	722 (4.4)	232 (3.9)	245 (3.9)	289 (4.7)	118 (6.3)
Education						
Lower qualification	154,780 (67.5)	12,388 (75.8)	4,440 (74.3)	4,855 (77.9)	4,645 (76.2)	1,456 (77.4)
Higher qualification	74,572 (32.5)	3,945 (24.2)	1,537 (25.7)	1,374 (22.1)	1,448 (23.8)	424 (22.6)
Average household income (£)						
< 18 000	63,595 (27.7)	6,712 (41.1)	2,457 (41.1)	3,063 (49.2)	2,276 (37.4)	725 (38.6)
18 000–30 999	58,603 (25.6)	4,461 (27.3)	1,632 (27.3)	1,666 (26.7)	1,661 (27.3)	534 (28.4)
31 000–51 999	55,537 (24.2)	3,086 (18.9)	1,127 (18.9)	945 (15.2)	1,279 (21.0)	364 (19.4)
52 000–100 000	41,230 (18.0)	1,699 (10.4)	622 (10.4)	466 (7.5)	719 (11.8)	207 (11.0)
> 100 000	10,387 (4.5)	375 (2.3)	139 (2.3)	89 (1.4)	158 (2.6)	50 (2.7)
BMI (kg/m^2)						
< 18.5	1,367 (0.6)	113 (0.7)	44 (0.7)	65 (1.0)	26 (0.4)	7 (0.4)
18.5–24.9	75,081 (32.7)	3,854 (23.6)	1,614 (27.0)	1,413 (22.7)	1,377 (22.6)	366 (19.5)
25–29.9	97,908 (42.7)	7,089 (43.4)	2,617 (43.8)	2,455 (39.4)	2,781 (45.6)	830 (44.1)
≥ 30	54,996 (24.0)	5,277 (32.3)	1,702 (28.5)	2,296 (36.9)	1,909 (31.3)	677 (36.0)
MET minutes per week for moderate activity (minutes)[a]	360.0 [120.0, 960.0]	360.00 [80.0, 960.0]	400.0 [120.0, 1,080.0]	360.0 [60.0, 960.0]	360.0 [80.0, 960. 0]	360.0 [80.0, 1,000.0]
Regular physical activity						
Low	65,501 (28.6)	5,449 (33.4)	1,911 (32.0)	2,293 (36.8)	2,009 (33.0)	579 (30.8)
Moderate	88,619 (38.6)	5,901 (36.1)	2,184 (36.5)	2,246 (36.1)	2,175 (35.7)	707 (37.6)
High	75,232 (32.8)	4,983 (30.5)	1,882 (31.5)	1,690 (27.1)	1,909 (31.3)	594 (31.6)
Smoking status						
Never	126,773 (55.3)	7,126 (43.6)	2,784 (46.6)	2,374 (38.1)	2,664 (43.7)	845 (44.9)
Former	78,318 (34.1)	6,405 (39.2)	2,279 (38.1)	2,608 (41.9)	2,324 (38.1)	750 (39.9)
Current	24,261 (10.6)	2,802 (17.2)	914 (15.3)	1,247 (20.0)	1,105 (18.1)	285 (15.2)
Alcohol consumption (times/week)						
Never	69,237 (30.2)	5,461 (33.4)	1,929 (32.3)	2,193 (35.2)	2,009 (33.0)	676 (36.0)
1–2	60,141 (26.2)	3,998 (24.5)	1,416 (23.7)	1,435 (23.0)	1,568 (25.7)	488 (26.0)
3–4	53,673 (23.4)	3,346 (20.5)	1,285 (21.5)	1,166 (18.7)	1,299 (21.3)	356 (18.9)
≥ 5	46,301 (20.2)	3,528 (21.6)	1,347 (22.5)	1,435 (23.0)	1,217 (20.0)	360 (19.1)
Vegetable consumption (servings/day)						
< 2.0	15,381 (6.7)	1,442 (8.8)	459 (7.7)	621 (10.0)	566 (9.3)	153 (8.1)
≥ 2.0–3.9	66,208 (28.9)	4,705 (28.8)	1,729 (28.9)	1,778 (28.5)	1,771 (29.1)	535 (28.5)
≥ 4.0–5.9	76,860 (33.5)	5,269 (32.3)	1,919 (32.1)	1,982 (31.8)	1,996 (32.8)	608 (32.3)
≥ 6.0	70,903 (30.9)	4,917 (30.1)	1,870 (31.3)	1,848 (29.7)	1,760 (28.9)	584 (31.1)
Fruit consumption (servings/day)						
< 2.0	75,917 (33.1)	6,059 (37.1)	2,113 (35.4)	2,367 (38.0)	2,332 (38.3)	691 (36.8)
≥ 2.0–2.9	58,226 (25.4)	3,897 (23.9)	1,470 (24.6)	1,462 (23.5)	1,447 (23.7)	434 (23.1)
≥ 3.0–3.9	44,444 (19.4)	2,917 (17.9)	1,102 (18.4)	1,054 (16.9)	1,085 (17.8)	335 (17.8)
≥ 4.0	50,765 (22.1)	3,460 (21.2)	1,292 (21.6)	1,346 (21.6)	1,229 (20.2)	420 (22.3)
CRP (mg/L)[a]	1.33 [0.66, 2.77]	1.83 [0.91, 3.77]	1.67 [0.83, 3.47]	2.15 [1.03, 4.44]	1.80 [0.92, 3.65]	1.86 [0.94, 3.79]
HbA1c (mmol/L)[a]	35.2 [32.7, 37.8]	36.5 [33.8, 39.6]	36.2 [33.6, 39.2]	37.1 [34.2, 41.0]	36.5 [33.9, 39.5]	36.50 [33.8, 39.90]

Table 1 (continued)

Characteristics	Overall (N = 229,352)	Incident MACE (N = 16,333)	Incident Stroke (N = 5,977)	Incident CVD death (N = 6,229)	Incident Myocardial Infarction (N = 6,093)	Incident Unstable angina (N = 1,880)
TC (mmol/L)[a]	5.68 [4.96, 6.45]	5.60 [4.78, 6.45]	5.59 [4.81, 6.41]	5.39 [4.55, 6.24]	5.78 [4.92, 6.62]	5.55 [4.70, 6.44]
HDL-C (mmol/L)[a]	1.41 [1.18, 1.68]	1.29 [1.09, 1.54]	1.34 [1.12, 1.61]	1.28 [1.07, 1.56]	1.25 [1.06, 1.48]	1.25 [1.06, 1.49]
LDL-C (mmol/L)[a]	3.55 [2.98, 4.14]	3.53 [2.89, 4.19]	3.50 [2.90, 4.11]	3.35 [2.71, 4.00]	3.69 [3.03, 4.36]	3.49 [2.84, 4.20]
TG (mmol/L)[a]	1.48 [1.04, 2.14]	1.70 [1.20, 2.44]	1.60 [1.13, 2.29]	1.66 [1.17, 2.36]	1.83 [1.29, 2.60]	1.85 [1.30, 2.60]
SBP (mmHg)[a]	137.0 [125.0, 150.0]	144.0 [132.0, 157.0]	144.0 [132.0, 158.0]	145.0 [132.0, 158.0]	144.0 [132.0, 157.0]	142.0 [130.0, 154.0]
DBP (mmHg)[a]	82.0 [76.0, 89.0]	84.00 [77.0, 91.0]	84.00 [77.0, 91.0]	84.0 [76.0, 91.0]	85.0 [78.0, 92.0]	83.0 [77.0, 91.0]
Waits circumference (cm)[a]	90.0 [80.0, 99.0]	95.00 [86.0, 104.0]	93.0 [84.0, 102.0]	97.0 [87.0, 107.0]	95.0 [87.0, 103.0]	96.0 [87.0, 104.0]
Hip circumference (cm) [a]	102.0 [97.0, 108.0]	103.0 [98.0, 109.0]	103.0 [98.0, 108.0]	104.0 [98.0, 111.0]	103.0 [98.0, 109.0]	104.0 [99.0, 110.0]
BMI (kg/m^2)[a]	26.71 [24.13, 29.83]	27.85 [25.10, 31.24]	27.37 [24.72, 30.52]	28.27 [25.16, 32.10]	27.89 [25.26, 31.04]	28.44 [25.60, 31.95]
Drug use						
Cholesterol lowering medication	34,957 (15.2)	4,666 (28.6)	1,564 (26.2)	2,186 (35.1)	1,598 (26.2)	693 (36.9)
Anti-hypertensive drug	44,574 (19.4)	5,811 (35.6)	2,045 (34.2)	2,756 (44.2)	1,942 (31.9)	706 (37.6)
Insulin	2,283 (1.0)	507 (3.1)	157 (2.6)	312 (5.0)	180 (3.0)	66 (3.5)
Aspirin	27,197 (11.9)	3,682 (22.5)	1,277 (21.4)	1,718 (27.6)	1,260 (20.7)	516 (27.4)
Non-aspirin NSAIDs	69,207 (30.2)	4,938 (30.2)	1,780 (29.8)	1,841 (29.6)	1,839 (30.2)	649 (34.5)
Disease history						
HTN	113,287 (49.4)	11,480 (70.3)	4,170 (69.8)	4,698 (75.4)	4,196 (68.9)	1,318 (70.1)
T2DM	12,403 (5.4)	2,169 (13.3)	699 (11.7)	1,144 (18.4)	759 (12.5)	266 (14.1)
CVD family history	122,229 (53.3)	112,654 (52.9)	3,470 (58.1)	3,515 (56.4)	3,515 (56.4)	1,205 (64.1)

BMI body mass index, *MET* metabolic equivalent task, *CRP* c-reactive protein, *HbA1c* glycated hemoglobin, *TC* total cholesterol, *HDL-C* HDL cholesterol, *LDL-C* LDL cholesterol, *TG* triglycerides, *SBP* systolic blood pressure, *DBP* diastolic blood pressure, *NSAID* nonsteroidal anti-inflammatory drug, *HTN* hypertension, *T2DM* type 2 diabetes mellitus, *CVD* cardiovascular disease

[a]Data are presented as median (Q1-Q3); otherwise, values were are presented as numbers (%)

HDL (M-HDL-PL) and BMI were found to be within the top 50 in importance. Cox proportional hazards models, adjusted for age, sex and BMI, demonstrated that each 1-standard deviation (SD) increase in Cystatin C, HbA1c, GGT, CRP and ALP were significantly associated with an increased risk of new-onset MACE, with the exception of creatinine, GlycA, and urea, which showed no significant association with incident hemorrhagic stroke risk. Specifically, for Cystatin C, each 1-SD increase was associated with a 0.14 (95% CI 0.12–0.15), 0.11 (95% CI 0.09–0.14), 0.13 (95% CI 0.12–0.14), 0.10 (95% CI 0.07–0.13), and 0.17 (95% CI 0.16–0.18) increase in the risk of ischemic stroke, hemorrhagic stroke, myocardial infarction, unstable angina, and CVD mortality, respectively. GlycA was significantly associated with new-onset ischemic stroke, myocardial infarction, unstable angina, and CVD mortality, where each 1-SD increase corresponded to a 0.10 (95% CI 0.07–0.14), 0.30 (95% CI 0.27–0.34), 0.19 (95% CI 0.14–0.25), and 0.24 (95% CI 0.21–0.27) increase in risk, respectively. In contrast, DHA exhibited a protective effect against MACE, especially in reducing the risk of CVD mortality. Specifically, each 1-SD increase in DHA was associated with a 20% decrease in CVD mortality risk (HR = 0.80, 95% CI 0.78–0.83). Notably, HDL-C and M-HDL-PL were more strongly associated with a reduced risk of new-onset myocardial infarction and unstable angina. For each 1-SD increase, HDL-C was associated with a 26% reduction in the risk of myocardial infarction (HR = 0.74, 95% CI 0.72–0.76) and a 29% reduction in the risk of unstable angina (HR = 0.71, 95% CI 0.67–0.76). Similarly, each 1-SD increase in M-HDL-PL was associated with a 17% reduction in myocardial infarction risk (HR = 0.83, 95% CI 0.81–0.86) and a 15% reduction in unstable angina risk (HR = 0.85, 95% CI 0.80–0.89) (Fig. 4).

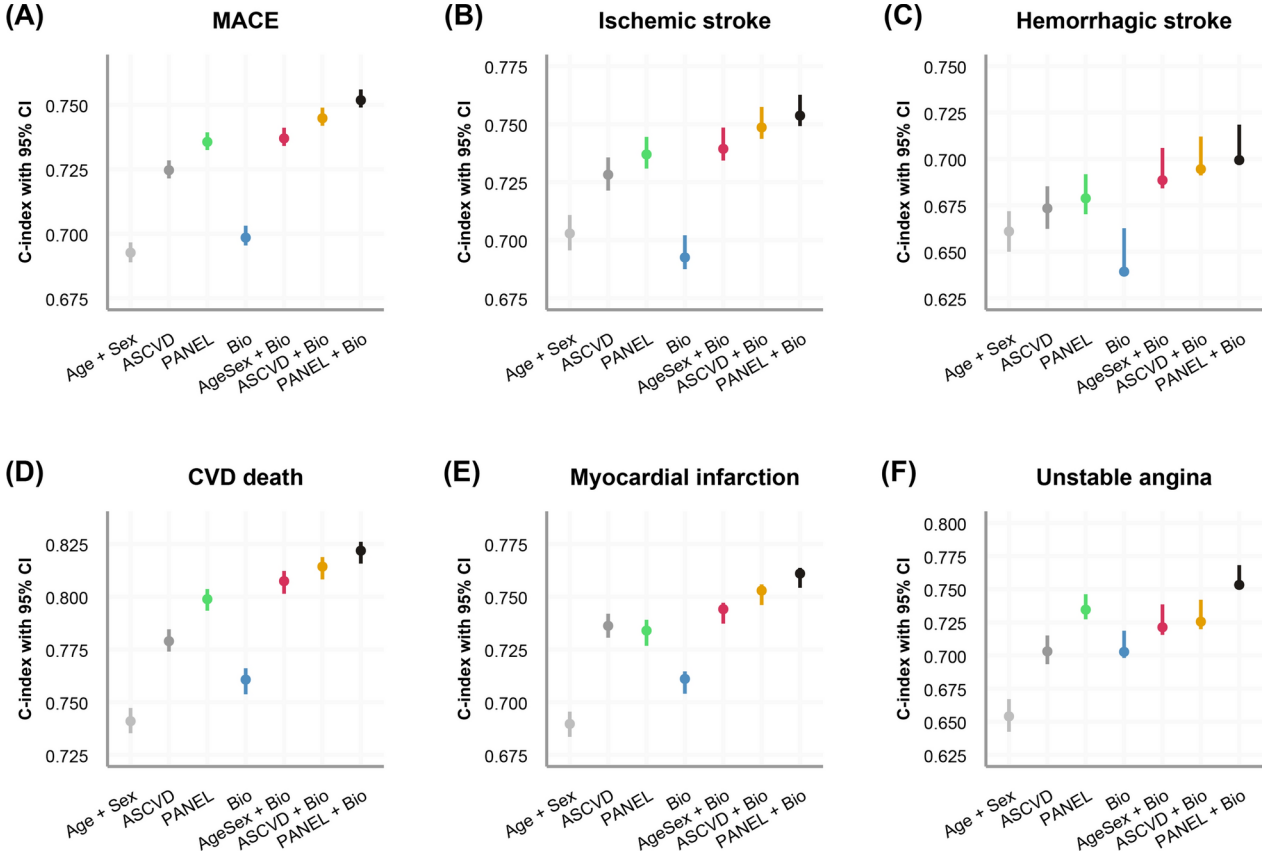


Fig. 2 Superior performance of combined biomarkers in cardiovascular risk stratification. This figure illustrates the comparative discriminative performance of incident MACE and its individual components, as assessed using Cox proportional hazards models. These models were developed based on selected biomarkers and three clinical predictor sets (Age + Sex, ASCVD, and PANEL) and their combinations. *MACE* major adverse cardiovascular events, *CVD* cardiovascular disease, *CI* confidence interval, *Bio* a set of 26 biochemical markers combined with Cor0.95 of Nonov Met, *Cor0.95* Spearman correlation coefficients less than 0.95, *Nonov Met*, high-throughput nuclear magnetic resonance (NMR) metabolic markers, excluding those that overlap with clinical biochemical markers

Elevated biomarkers and their association with incident MACE and cardiovascular outcomes

In this study, we identified several biomarkers associated with an increased risk of MACE and their individual components (Fig. 5). Baseline factors, such as age ≥ 60 years, male, cystatin C ≥ 1.0 mg/L, HbA1c ≥ 39.0 mmol/mol, GGT ≥ 50 U/L, and CRP ≥ 3.0 mg/L were each significantly linked to higher risks for MACE, ischemic stroke, hemorrhagic stroke, cardiovascular death, myocardial infarction, and unstable angina. Notably, elevated Cystatin C levels (≥ 1.0 mg/L) were linked to increased risks across all cardiovascular outcomes, with HRs ranging from 1.25 to 1.97 depending on the event. Similarly, HbA1c levels ≥ 39.0 mmol/mol were associated with increased risks of MACE, ischemic stroke, cardiovascular death, myocardial infarction, and unstable angina, with HRs ranging from 1.35 to 1.75, as well as an 18% higher risk of hemorrhagic stroke. Elevated GGT levels (≥ 50 U/L) were associated with higher risks for all the aforementioned outcomes, with HRs ranging from 1.21 to 1.45, while CRP levels ≥ 3.0 mg/L were similarly linked to increased risks, with HRs ranging from 1.21 to 1.72. Furthermore, individuals with GlycA concentrations above 0.85, 0.95, 0.85, 0.80, and 0.825 mmol/L had significantly higher risks of incident MACE, ischemic stroke, cardiovascular death, myocardial infarction, and unstable angina, with HRs of 1.34 (1.30–1.39), 1.29 (1.19–1.40), 1.41 (1.34–1.48), 1.57 (1.49–1.66), and 1.30 (1.18–1.42), respectively, compared to those with lower GlycA concentrations.

On the other hand, IGF-1 levels ≥ 18.0 nmol/L were associated with reduced risks for ischemic stroke, hemorrhagic stroke, cardiovascular death and unstable angina, with HRs of 0.75 to 0.85, suggesting a protective effect. Moreover, higher omega-3 levels (≥ 0.5 mmol/L) were linked to a reduced risk of hemorrhagic stroke (HR: 0.80; 95% CI 0.73–0.89) and cardiovascular mortality (HR: 0.71; 95% CI 0.68–0.75), while elevated DHA levels (≥ 0.2 mmol/L) were associated with a decreased risk of ischemic stroke (HR: 0.85; 95% CI 0.79–0.90) and hemorrhagic stroke (HR: 0.85; 95% CI 0.76–0.94).

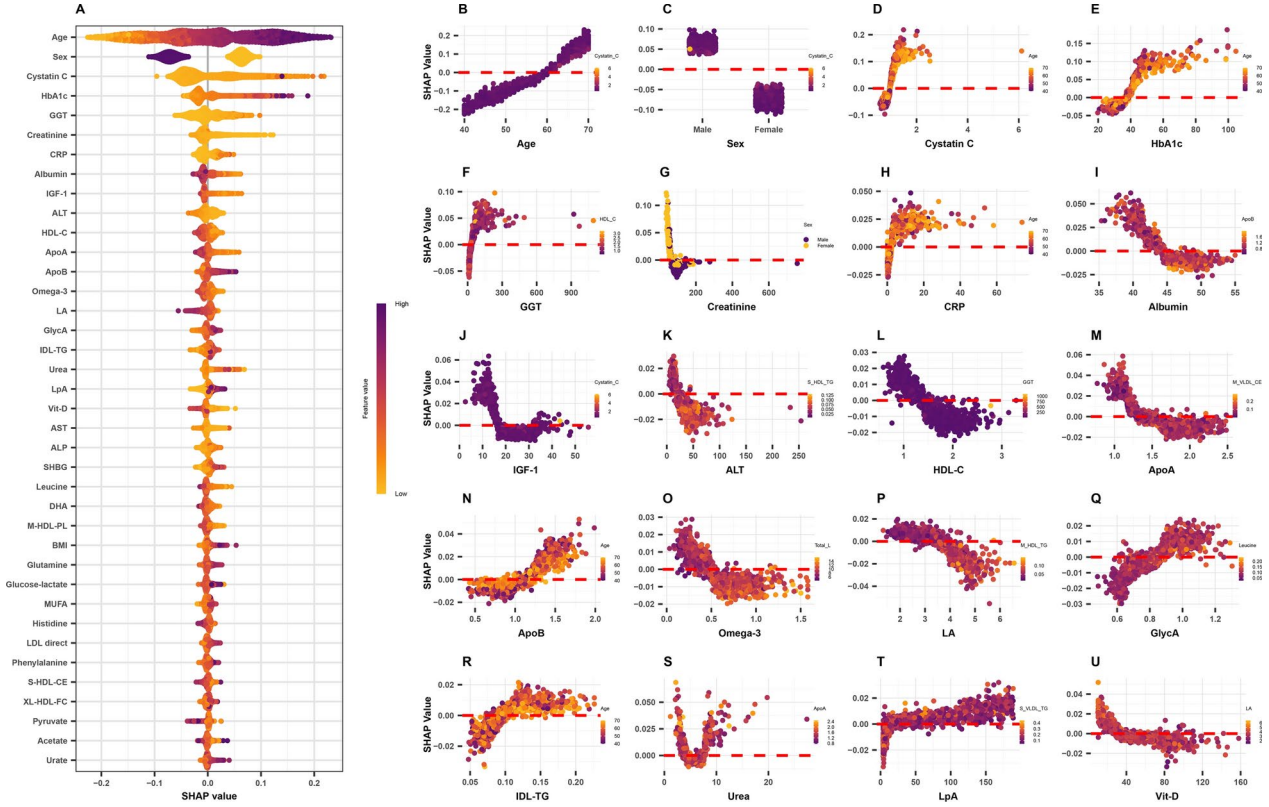


Fig. 3 SHAP analysis reveals key factors influencing the prediction of MACE (**A**). SHAP importance plot for the top 50% features in onset MACE prediction; (**B**–**U**). SHAP dependence plot for top 20 features in onset MACE prediction. *MACE* major adverse cardiovascular events, *SHAP* SHapley Additive exPlanations

RCS-Cox analysis showed a nonlinear association between BMI and incident MACE. Specifically, a U-shaped relationship was observed between BMI and CVD-related mortality risk (Supplementary Fig. 8).

Enhanced cardiovascular risk stratification using baseline risk index and score

We further assessed the cumulative hazard for each cardiovascular endpoint by stratifying participants into three groups based on quintiles of the BRI: the bottom 20%, middle 20%, and top 20%. As shown in Fig. 6A, there was a clear, significant increase in the cumulative risk of new-onset MACE and its individual components with higher BRI. Participants in the top 20% had substantially higher cumulative risks across all types of MACE, with particularly pronounced differences observed for cardiovascular death and myocardial infarction. These findings underscore the potential of the BRI as a robust tool for cardiovascular risk stratification.

Figure 6B further illustrates the classification of low-, medium-, and high-risk groups based on tertiles of the BRS for MACE and its components. For each group, we report a series of absolute risk indicators, including new-onset cases, incidence proportion, and ID, as well as relative risk indicators. Relative risk is presented using multivariable-adjusted HR with 95% CI, with the low-risk group as the reference. For MACE, the incidence proportion was 3.37%, 6.56% and 12.50%, and the ID was 2.59, 4.99, and 10.00 per 1,000 person-years for the low-, moderate-, and high-risk groups, respectively. For ischemic stroke and hemorrhagic stroke, the ID increased from 0.74 to 2.56 and from 0.33 to 0.92 per 1,000 person-years, respectively, from the low-risk group to the high-risk group. Similarly, for CVD death, myocardial infarction, and unstable angina, the incidence proportion increased from 1.03% to 6.48%, 1.05% to 4.88%, and 0.37% to 1.48%, respectively. The ID increased from 0.76 to 4.98, 0.78 to 3.77, and 0.28 to 1.13 per 1,000 person-years, respectively, from the low-risk group to the high-risk group. Additionally, the results show a significant upward trend in IDs across the risk groups, from low to medium to high, for all cardiovascular outcomes. Notably, the BRS exhibited the strongest risk stratification effects for cardiovascular death, new-onset myocardial infarction, and unstable angina. Compared to the low-risk group, the HRs for the medium-risk group increased by more than 35%, while the high-risk group showed relative risks of 2.86 (95% CI 2.65–3.08), 2.32 (95% CI 2.14–2.53), and 2.20 (95% CI 1.91–2.53) for these outcomes, respectively. The stratification for ischemic stroke showed relative

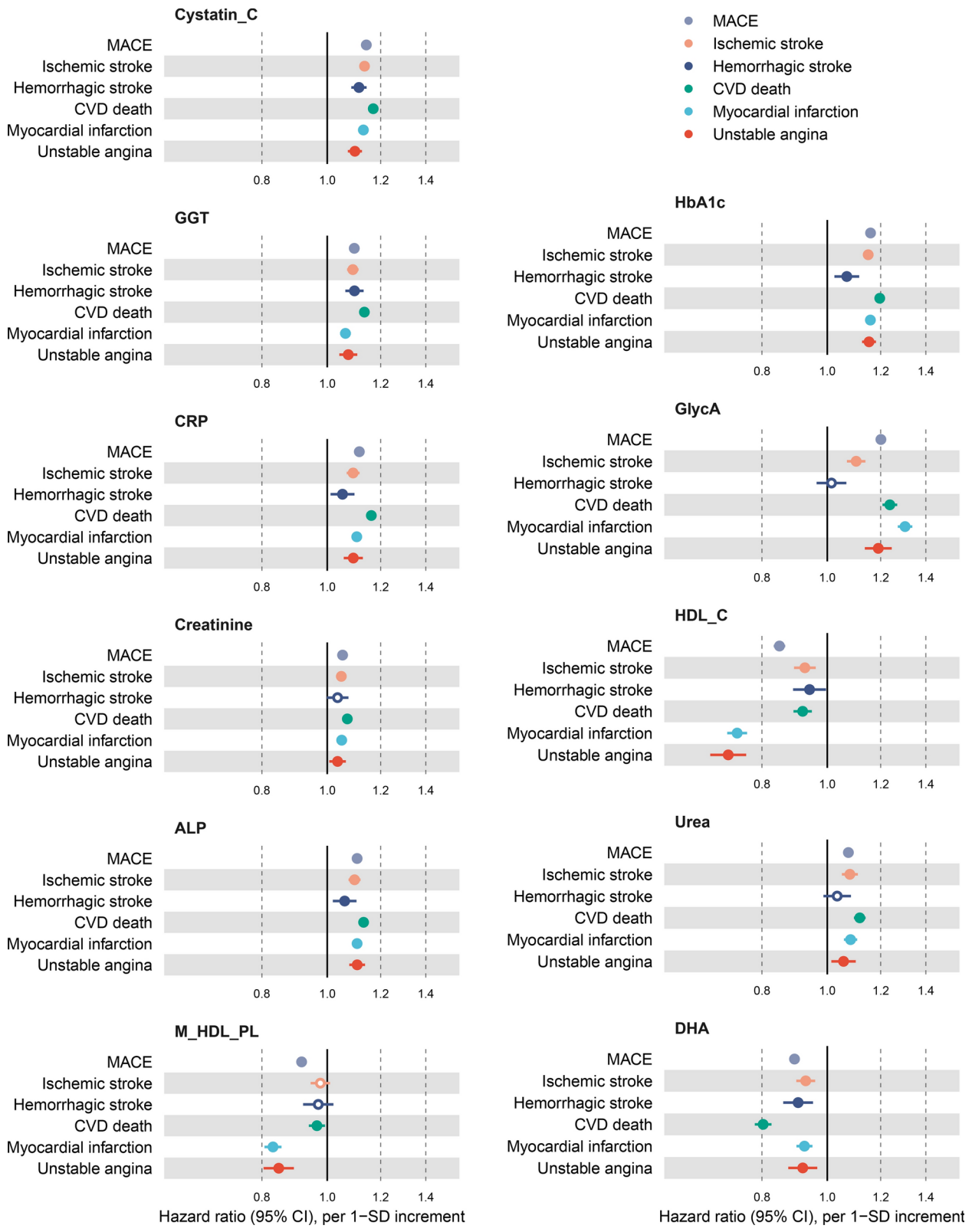


Fig. 4 Cox proportional hazards model identifies key biomarkers associated with new-onset MACE and its components. *MACE* major adverse cardiovascular events, *CVD* cardiovascular disease, *CI* confidence interval, *SD* standard deviation, *GGT* gamma-glutamyltransferase, *HbA1c* glycated hemoglobin, *CRP* c-reactive protein, *GlycA* glycoprotein acetyls, *HDL-C* HDL cholesterol, *ALP* alkaline phosphatase, *M-HDL-PL* phospholipids in medium HD, *DHA* docosahexaenoic acid. *Note*: Cox proportional hazards regression adjusted for age, sex and body mass index

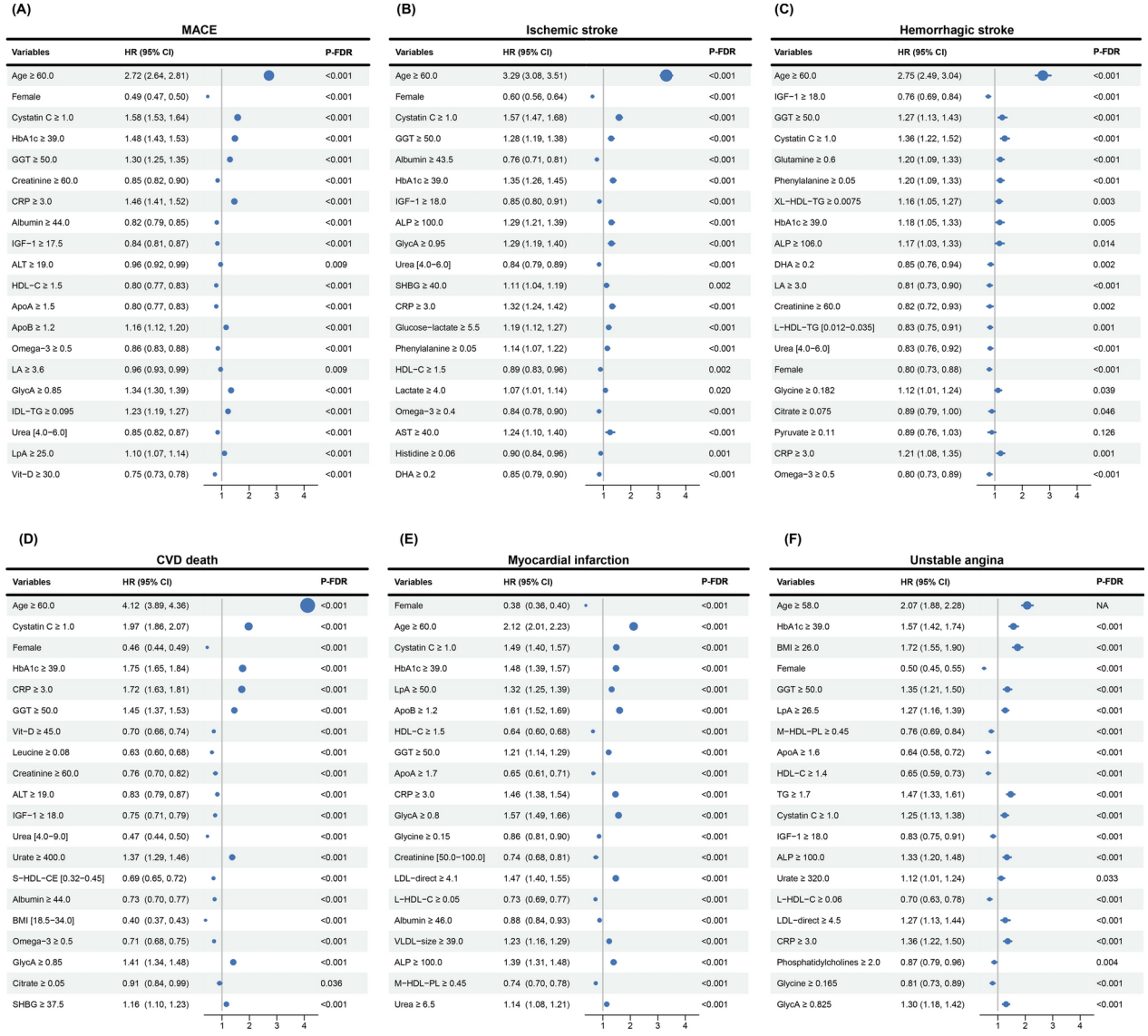
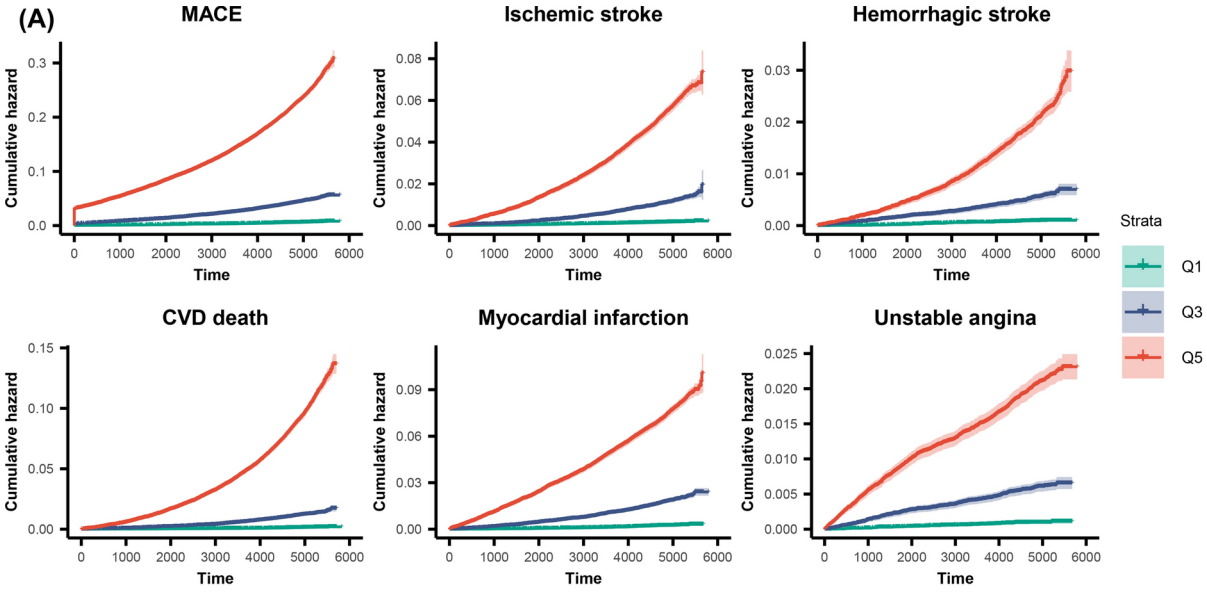


Fig. 5 Elevated biomarkers and their association with incident MACE and cardiovascular outcomes. The HR and 95% CI for the association between the elevated biomarkers and new-onset MACE (**A**), ischemic stroke (**B**), hemorrhagic stroke (**C**), CVD mortality (**D**), myocardial infarction (**E**), and unstable angina (**F**). *MACE* major adverse cardiovascular events, *CVD* cardiovascular disease, *HR* hazard ratio, *CI* confidence interval. *Note*: Cox proportional hazards regression adjusted for age, sex and body mass index

risks of 1.21 (95% CI 1.11–1.32) in the moderate-risk group and 1.64 (95% CI 1.50–1.78) in the high-risk group. The stratification effect of the BRS was somewhat weaker for hemorrhagic stroke, with relative risks of 1.40 (95% CI 1.23–1.59) and 1.58 (95% CI 1.56–2.03) for the medium- and high-risk groups, respectively.

Discussion

A key methodological feature of this study is the integration of the CatBoost model with SHAP analysis, which facilitates the quantification of variable importance and the identification of approximate optimal binary thresholds [16, 22, 37, 41–43]. SHAP analysis further enhances model interpretability by visualizing each predictor's contribution and importance ranking, as well as estimating the optimal binary thresholds for each variable and illustrating their relationship with predicted probabilities (e.g., age ≥ 60 years, cystatin C ≥ 1.0 mg/L). This innovative approach addresses the limitations of "black-box" models, holding promise for enhancing cardiovascular risk prediction. To the best of our knowledge, this study is the first to report the importance ranking and corresponding binary cutoff values of chemical and metabolic biomarkers for predicting incident MACE and its components. This enables the identification of modifiable risk or protective factors that significantly influence the onset of the outcomes of interest and achieves the following goals: (1) risk stratification and classification of



(B)

Subgroup	BRS	Total No. of participants	No. of cases (%)	ID	HR (95% CI)	P-FDR
MACE						
Low risk	[0, 6]	91,929	3,188 (3.37)	2.59	1.00 (reference)	
Moderate risk	(6, 8]	67,847	4,450 (6.56)	4.99	1.35 (1.29–1.42)	<0.001
High risk	(8, 20]	69,576	69,576 (12.50)	10.00	1.88 (1.79–1.96)	<0.001
Ischemic stroke						
Low risk	[0, 5]	100,397	1,006 (1.00)	0.74	1.00 (reference)	
Moderate risk	(5, 7]	64,476	1,116 (1.73)	1.30	1.21 (1.11–1.32)	<0.001
High risk	(7, 20]	64,479	2,119 (3.27)	2.56	1.64 (1.50–1.78)	<0.001
Hemorrhagic stroke						
Low risk	[0, 6]	93,355	418 (0.45)	0.33	1.00 (reference)	
Moderate risk	(6. 8]	76,834	595 (0.77)	0.58	1.40 (1.23–1.59)	<0.001
High risk	(8, 20]	59,163	706 (1.19)	0.92	1.58 (1.56–2.03)	<0.001
CVD death						
Low risk	[0, 5]	103,105	1,065 (1.03)	0.76	1.00 (reference)	
Moderate risk	(5, 7]	74,090	1,783 (2.41)	1.78	1.54 (1.44–1.68)	<0.001
High risk	(7, 20]	52,157	3,881 (6.48)	4.98	2.86 (2.65–3.08)	<0.001
Myocardial infarction						
Low risk	[0, 5]	87,054	913 (1.05)	0.78	1.00 (reference)	
Moderate risk	(5, 8]	71,413	1,722 (2.41)	1.82	1.53 (1.41–1.661)	<0.001
High risk	(8, 20]	70,885	3,458 (4.88)	3.77	2.32 (2.14–2.53)	<0.001
Unstable angina						
Low risk	[0, 6]	93,988	349 (0.37)	0.28	1.00 (reference)	
Moderate risk	(6, 9]	59,014	402 (0.68)	0.51	1.35 (1.17–1.57)	<0.001
High risk	(9, 20]	76,350	1,129 (1.48)	1.13	2.20 (1.91–2.53)	<0.001

Fig. 6 (See legend on next page.)

(See figure on previous page.)

Fig. 6 Enhanced cardiovascular risk stratification using baseline risk index and biomarker risk score. (**A**). Cumulative risk of incident MACE and its components during follow-up, stratified by biomarker risk index; (**B**). Estimation of absolute and relative risk for new-onset MACE and its components, stratified by biomarker risk score, using the multivariable-adjusted Cox proportional hazards model. *MACE* major adverse cardiovascular events, *CVD* cardiovascular disease, *BRS* biomarker risk score, *HR* hazard ratio, *CI* confidence interval, *ID* incidence density. *Notes* The number of new-onset cases and the incidence proportion were presented as No. of cases (%); ID are provided per 1000 person-years; Cox proportional hazards regression adjusted for age, sex, education, average household income, ethnicity, smoke status, drinking frequency, met minutes for moderate activity per week, total fruit intake, total vegetable consumption, systolic blood pressure, diastolic blood pressure, blood pressure medication, insulin, cholesterol lowering medication, a family history of cardiovascular disease, prevalent type 2 diabetes mellitus, and prevalent hypertension

the population based on the estimated binary thresholds for risk and protective factors, facilitating more precise management of CVD; and (2) recognition of intervention targets, such as reducing risk factors below the corresponding binary threshold or enhancing protective factors above it through interventions, supplements, prebiotics, or probiotics.

Additionally, multiple variable selection strategies, including AUC, JMIM, and correlation analysis, were employed. In determining the optimal predictor subset, we accounted for not only the predictive performance of the model (C-index) but also the number of variables and the model's complexity. Our parsimonious models yielded a C-index of 0.752 (95% CI 0.749–0.756) for MACE incidence, demonstrating comparable performance to the full model, which combined all measured overlapping metabolic biomarkers, the selected 26 clinical chemistry biomarkers, and the defined PANEL information (C-index: 0.753; 95% CI 0.751–0.758). Overall, the final predictor subset, "PANEL + All Bioche + Cor0.95 of Nonov Met," achieved a satisfactory balance between predictive accuracy and model simplicity.

The study explored the improvement in discriminatory performance achieved by biomarkers through comparisons with traditional predictive sets, such as Age + Sex, ASCVD, and PANEL. While conventional cardiovascular risk assessment tools predominantly rely on demographic characteristics and lipid-related indicators, our study incorporated a broader range of biomarkers from diverse metabolic pathways, such as cystatin C, GGT, GlycA, and DHA, allowing for the evaluation of additional modifiable risk factors. Notably, we observed heterogeneity in the optimal binary thresholds of key variables across different disease endpoints. For instance, the binary thresholds for age, GlycA, IGF-1, and omega-3 varied between risk stratifications for MACE and its components. These findings provide a valuable foundation for developing personalized interventions tailored to specific cardiovascular outcomes.

By integrating an extensive range of biomarkers with traditional CVD risk factors, this study proposes a more comprehensive framework for CVD risk assessment, facilitating the optimization of personalized prevention strategies. For example, individuals with cystatin C, GlycA, or HbA1c levels exceeding specific thresholds could be prioritized for targeted interventions. Furthermore, these findings support public health policymakers in stratifying moderate- or high-risk populations, thereby improving the allocation of healthcare resources. If the biomarkers used to generate BRI and BRS are routinely measured in clinical practice in the near future, they may offer a promising tool for the precise identification and early intervention of high-risk populations. Unfortunately, some of these biomarkers are not currently part of routine clinical testing. The widespread application of this multi-biomarker-based cardiovascular prediction and risk stratification strategy in clinical practice will depend on the future accessibility, validation, and standardization of these biomarkers.

This study underscores the critical role of specific biomarkers in predicting and stratifying risk for MACE through data analysis and machine learning algorithms. These biomarkers are biologically relevant, playing significant roles regulating physiological functions associated with CVD. Rather than relying exclusively on statistical significance, we explore several potential biological mechanisms that may explain the associations between biomarkers and CVD outcomes. These mechanisms are rooted in well-established biological pathways, which lend biological plausibility to the observed associations and enhance their clinical relevance. Cystatin C is linked to elevated monocyte counts that contribute to plaque formation in arterial walls; high levels of Cystatin C often indicate renal dysfunction, which is associated with increased carotid intima-media thickness and heightened atherosclerosis risk [7, 8, 44, 45]. GlycA acts as a vital marker of chronic inflammation, reflecting the overall systemic inflammatory burden and playing a key role in the pathogenesis of various diseases, including cardiovascular events. As a composite biomarker, GlycA is particularly sensitive to fluctuations in the inflammatory response, capturing subtle alterations in systemic inflammation that may not be detected by traditional markers such as CRP. This unique sensitivity allows GlycA to identify individuals at elevated cardiovascular risk more effectively, especially when assessed alongside other established risk factors [10, 46]. HbA1c, a marker of long-term glucose control, is strongly correlated with the onset and progression of CVD, as chronic hyperglycemia leads to oxidative stress, resulting in endothelial dysfunction and atherosclerosis through advanced glycation end-products (AGEs) [11, 47, 48]. IGF-1 plays

a key role in cellular growth and metabolism, demonstrating anti-inflammatory properties modulated by pro-inflammatory cytokines [49–51]. Several previous studies suggested that elevated GGT levels contribute to prooxidant activity, leading to oxidative damage at the cellular and molecular levels, including tissue and DNA damage, which can play a crucial role in cardiovascular disease and other chronic conditions [52–54]. Additionally, DHA, an omega-3 fatty acid, influences various risk factors, including triglyceride levels and inflammation, thereby highlighting its protective effects against CVD [19, 55, 56]. These insights further validate that our machine learning model has accurately identified these key biomarkers related to MACE, demonstrating their scientific significance in prediction and risk stratification.

In conclusion, this study demonstrates the potential of integrating traditional risk factors with biochemical and metabolomic biomarkers to enhance prediction and risk stratification for new-onset MACE and its components. By employing advanced machine learning models and SHAP analysis, we identified key biomarkers with significant predictive power, including Cystatin C, HbA1c, GlycA, and GGT, which can facilitate the early identification of high-risk individuals. These findings suggest that a multi-biomarker approach could refine CVD risk evaluation and guide targeted prevention strategies, with possible implications for clinical practice and public health, pending further validation and standardization.

Limitations

Although this study benefits from several strengths, including a large-scale population sample, an extended follow-up duration, comprehensive clinical biochemical measurements, and high-throughput metabolomics data that enabled the replication of previous findings and the discovery of novel blood biomarkers, there are notable limitations that should be considered in the interpretation of the results.

Firstly, the prediction models were developed using a population aged 40 to 69 years, limiting their generalizability to older individuals, who have a notably higher risk of incident CVD. Future research should include a broader age range, particularly older populations, to enhance the generalizability of these models and improve their applicability in high-risk groups. Additionally, the study sample predominantly comprised individuals of European descent from the UK Biobank, and the demographic, lifestyle, and health-related characteristics of UK Biobank participants may differ from those of other populations, restricting the external validity of our findings. This makes it difficult to directly generalize our results to populations with different racial, health behavior, or socio-economic characteristics. Future studies should include more diverse populations to validate and extend our findings. Another limitation of this study is the absence of an external validation cohort. Without validation in an independent population, the generalizability and robustness of the proposed models remain uncertain. While our findings are promising, the applicability of the models to other populations needs to be further confirmed. Future studies should focus on validating our results in diverse, independent cohorts to assess the external validity of our model and its performance across different demographic and clinical groups. Fourth, while this innovative approach addresses the limitations of'black-box' models and holds promise for enhancing cardiovascular risk prediction, we recognize that several biomarkers identified and further used in the BRI and BRS proposed in this study are not routinely measured in clinical laboratories, which limits the immediate applicability and ease of use of these scores in clinical settings at present. Fifth, the study did not provide an estimation of absolute risk for individual biomarkers, which limits the clinical applicability of the findings. Including absolute risk calculations would enhance the practical utility of the results, as relative risk measures alone may not fully convey the real-world impact of the identified associations. Furthermore, while the UK Biobank offers data on 170 circulating metabolites, it does not provide a complete representation of the metabolome, with a predominant focus on metabolites related to lipoproteins and lipids. This limitation constrains our ability to investigate the potential contributions of other metabolite classes, such as amino acids and fatty acids, to CVD risk prediction. Notably, the estimation of binary thresholds for biomarkers used to predict various types of MACE onset may be considered relatively coarse. Nevertheless, the thresholds established for HbA1c and HDL-C are largely consistent with current clinical guidelines, partially reinforcing the reliability of our binary classification method. Finally, although we have proposed approximate binary thresholds for various biomarkers, the implementation of these metabolite thresholds in real-world settings must take into account factors such as variations in testing methodologies, which could influence metabolite concentration measurements. In summary, while this study presents important findings, future research should focus on addressing these limitations to enhance the applicability and generalizability of the results.

Abbreviations

MACE	Major adverse cardiovascular events
CVD	Cardiovascular disease
NMR	Nuclear magnetic resonance
AUC	Area under the receiver operating characteristic curve
JMIM	Minimal joint mutual information maximization
ASCVD	Cardiovascular predictors set
BRI	Biomarker risk index
BRS	Biomarker risk score
HR	Hazard ratio
CI	Confidence interval

ID	Incidence density
SD	Standard deviation
IQR	Interquartile range
SHAP	SHapley Additive exPlanations
ICD-9	International Classification of Diseases, 9th Revision
ICD-10	International Classification of Diseases, 10th Revision
T2DM	Type 2 diabetes mellitus
BMI	Body mass index
GlyA	Glycoprotein acetyls
CRP	C-reactive protein
DBP	Diastolic blood pressure
SBP	Systolic blood pressure
HbA1c	Glycated hemoglobin
IGF-1	Insulin-like growth factor 1
HDL	High density lipoprotein
M-HDL-PL	Phospholipids in medium HDL
TG	Triglycerides
DHA	Docosahexaenoic acid
GGT	Gamma glutamyltransferase
ALP	Alanine aminotransferase

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12933-025-02711-x.

Supplementary Material 1

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Author contributions
Study concept and design: HW Zhou, Z Li, and Y He, and XR Zhang. Study supervision: Z Li, and HW Zhou. Acquisition, analysis, or interpretation of data: XR Zhang, ZH Li, and WF Zhong. Contributed to discussion of the results: HW Zhou, Z Li, and XR Zhang. Drafting of the manuscript: XR Zhang. Critical revision of the manuscript for important intellectual content: All authors. Final approval of the article: All authors. Statistical expertise: XR Zhang, ZH Li, D Liu, PD Zhang, and WF Zhong. Funding acquisition: HW Zhou, Y He, XR Zhang, and J Gao. Technical, material or administrative support: RY Liu, JL Huang, and JX Fu.

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Availability of data and materials
The data used in this study were available from the UK Biobank with restrictions applied. The data were used under license. UK Biobank data are available to all researchers for health-related research and studies in the public interest, and access can be requested through their established protocol (https://www.ukbiobank.ac.uk/register-apply/).

Declarations

Ethics approval and consent to participate
This research complied with the principles of the Declaration of Helsinki. Written informed consent was obtained from all participants prior to their involvement, and ethical approval was granted by the North West Multi-Center Research Ethics Committee (reference: 11/NW/0382; https://www.ukbiobank.ac.uk/learn-more-about-ukbiobank/about-us/ethics). Any additional ethical approval was adjudged unnecessary for the present study.

Patient and public involvement
Patients and/or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

Patient consent for publication
Not required.

Competing interests
The authors declare no competing interests.

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References

1. Timmis A, Aboyans V, Vardas P, Townsend N, Torbica A, Kavousi M, et al. European Society of Cardiology: the 2023 Atlas of Cardiovascular Disease Statistics. Eur Heart J. 2024;45:4019–62.
2. Joynt Maddox KE, Elkind MSV, Aparicio HJ, Commodore-Mensah Y, de Ferranti SD, Dowd WN, et al. Forecasting the burden of cardiovascular disease and stroke in the United States through 2050-prevalence of risk factors and disease: a presidential advisory from the American Heart Association. Circulation. 2024;150:e65-88.
3. Buergel T, Steinfeldt J, Ruyoga G, Pietzner M, Bizzarri D, Vojinovic D, et al. Metabolomic profiles predict individual multidisease outcomes. Nat Med. 2022;28:2309–20.
4. SCORE2 working group and ESC Cardiovascular risk collaboration. SCORE2 risk prediction algorithms: new models to estimate 10-year risk of cardiovascular disease in Europe. Eur Heart J. 2021;42:2439–54.
5. Goff DC, Lloyd-Jones DM, Bennett G, Coady S, D'Agostino RB, Gibbons R, et al. 2013 ACC/AHA guideline on the assessment of cardiovascular risk: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. J Am Coll Cardiol. 2014;63:2935–59.
6. Honigberg MC, Zekavat SM, Pirruccello JP, Natarajan P, Vaduganathan M. Cardiovascular and kidney outcomes across the glycemic spectrum: insights from the UK biobank. J Am Coll Cardiol. 2021;78:453–64.
7. Shlipak MG, Sarnak MJ, Katz R, Fried LF, Seliger SL, Newman AB, et al. Cystatin C and the risk of death and cardiovascular events among elderly persons. N Engl J Med. 2005;352:2049–60.
8. Correa S, Morrow DA, Braunwald E, Davies RY, Goodrich EL, Murphy SA, et al. Cystatin C for risk stratification in patients after an acute coronary syndrome. J Am Heart Assoc. 2018;7:e009077.
9. Ridker PM, Moorthy MV, Cook NR, Rifai N, Lee I-M, Buring JE. Inflammation, Cholesterol, Lipoprotein(a), and 30-Year Cardiovascular Outcomes in Women. N Engl J Med. 2024;391:2087–97.
10. Joshi AA, Lerman JB, Aberra TM, Afshar M, Teague HL, Rodante JA, et al. GlycA is a novel biomarker of inflammation and subclinical cardiovascular disease in psoriasis. Circ Res. 2016;119:1242–53.
11. Li F-R, Zhang X-R, Zhong W-F, Li Z-H, Gao X, Kraus VB, et al. Glycated hemoglobin and all-cause and cause-specific mortality among adults with and without diabetes. J Clin Endocrinol Metab. 2019;104:3345–54.
12. Julkunen H, Cichońska A, Tiainen M, Koskela H, Nybo K, Mäkelä V, et al. Atlas of plasma NMR biomarkers for health and disease in 118,461 individuals from the UK Biobank. Nat Commun. 2023;14:604.
13. Chen J-X, Li Y, Zhang Y-B, Wang Y, Zhou Y-F, Geng T, et al. Nonlinear relationship between high-density lipoprotein cholesterol and cardiovascular disease: an observational and Mendelian randomization analysis. Metabolism. 2024;154:155817.

14. Wu J, Giles C, Dakic A, Beyene HB, Huynh K, Wang T, et al. Lipidomic risk score to enhance cardiovascular risk stratification for primary prevention. J Am Coll Cardiol. 2024;84:434–46.
15. Tikkanen E, Jägerroos V, Holmes MV, Sattar N, Ala-Korpela M, Jousilahti P, et al. Metabolic biomarker discovery for risk of peripheral artery disease compared with coronary artery disease: lipoprotein and metabolite profiling of 31 657 individuals from 5 prospective cohorts. J Am Heart Assoc. 2021;10:e021995.
16. Gou W, Ling C-W, He Y, Jiang Z, Fu Y, Xu F, et al. Interpretable machine learning framework reveals robust gut microbiome features associated with type 2 diabetes. Diabetes Care. 2021;44:358–66.
17. Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, et al. From local explanations to global understanding with explainable AI for trees. Nat Mach Intell. 2020;2:56–67.
18. Zhang P-D, Zhang X-R, Zhang A, Li Z-H, Liu D, Zhang Y-J, et al. Associations of genetic risk and smoking with incident COPD. Eur Respir J. 2022;59:2101320.
19. Li Z-H, Zhong W-F, Liu S, Kraus VB, Zhang Y-J, Gao X, et al. Associations of habitual fish oil supplementation with cardiovascular outcomes and all cause mortality: evidence from a large population based cohort study. BMJ. 2020;368:m456.
20. Lee JM, Koo B-K, Shin E-S, Nam C-W, Doh J-H, Hwang D, et al. Clinical implications of three-vessel fractional flow reserve measurement in patients with coronary artery disease. Eur Heart J. 2018;39:945–51.
21. Almuwaqqat Z, Hui Q, Liu C, Zhou JJ, Voight BF, Ho YL, et al. Long-term body mass index variability and adverse cardiovascular outcomes. JAMA Netw Open. 2024;7:e243062.
22. Li L, Prato CG, Wang Y. Ranking contributors to traffic crashes on mountainous freeways from an incomplete dataset: a sequential approach of multivariate imputation by chained equations and random forest classifier. Accid Anal Prev. 2020;146:105744.
23. van Buuren S, Groothuis-Oudshoorn K. mice: multivariate imputation by chained equations in R. J Stat Softw. 2011;45:1–67.
24. Bommert A, Welchowski T, Schmid M, Rahnenführer J. Benchmark of filter methods for feature selection in high-dimensional gene expression survival data. Brief Bioinform. 2022;23:bbab354.
25. Kulesa A, Krzywinski M, Blainey P, Altman N. Sampling distributions and the bootstrap. Nat Methods. 2015;12:477–8.
26. Aune D, Schlesinger S, Norat T, Riboli E. Body mass index, abdominal fatness, and the risk of sudden cardiac death: a systematic review and dose-response meta-analysis of prospective studies. Eur J Epidemiol. 2018;33:711–22.
27. Chen Y, Yu W, Lv J, Sun D, Pei P, Du H, et al. Early adulthood BMI and cardiovascular disease: a prospective cohort study from the China Kadoorie Biobank. Lancet Public Health. 2024;9:e1005–13.
28. Fernandes-Silva MM, Shah AM, Claggett B, Cheng S, Tanaka H, Silvestre OM, et al. Adiposity, body composition and ventricular-arterial stiffness in the elderly: the Atherosclerosis Risk in Communities Study. Eur J Heart Fail. 2018;20:1191–201.
29. Strazzullo P, D'Elia L, Cairella G, Garbagnati F, Cappuccio FP, Scalfi L. Excess body weight and incidence of stroke: meta-analysis of prospective studies with 2 million participants. Stroke. 2010;41:e418-426.
30. Zaccardi F, Dhalwani NN, Papamargaritis D, Webb DR, Murphy GJ, Davies MJ, et al. Nonlinear association of BMI with all-cause and cardiovascular mortality in type 2 diabetes mellitus: a systematic review and meta-analysis of 414,587 participants in prospective studies. Diabetologia. 2017;60:240–8.
31. Beyene HB, Giles C, Huynh K, Wang T, Cinel M, Mellett NA, et al. Metabolic phenotyping of BMI to characterize cardiometabolic risk: evidence from large population-based cohorts. Nat Commun. 2023;14:6280.
32. Global Burden of Metabolic Risk Factors for Chronic Diseases Collaboration (BMI Mediated Effects), Lu Y, Hajifathalian K, Ezzati M, Woodward M, Rimm EB, et al. Metabolic mediators of the effects of body-mass index, overweight, and obesity on coronary heart disease and stroke: a pooled analysis of 97 prospective cohorts with 1·8 million participants. Lancet. 2014;383:970–83.
33. Qi J, Yang R, Wang P. Application of explainable machine learning based on Catboost in credit scoring. In: Journal of Physics: Conference Series. 2021. vol. 1955, pp. 012039.
34. Dorogush AV, Ershov V, Gulin A. CatBoost: gradient boosting with categorical features support. http://arxiv.org/abs/1810.11363 (2018). Accessed 22 Feb 2025.
35. Prokhorenkova L, Gusev G, Vorobev A, Dorogush AV, Gulin A. CatBoost: unbiased boosting with categorical features. In: Proceedings of the 32nd International Conference on Neural Information Processing Systems. Red Hook: Curran Associates Inc.; 2018. p. 6639–49.
36. Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. In: Proceedings of the 31st International Conference on Neural Information Processing Systems. Red Hook: Curran Associates Inc.; 2017. p. 4768–77.
37. Yan Y, Bao X, Chen B, Li Y, Yin J, Zhu G, et al. Interpretable machine learning framework reveals microbiome features of oral disease. Microbiol Res. 2022;265:127198.
38. Szklo M, Nieto FJ. Epidemiology: beyond the basics. 3rd ed. Burlington: Jones & Bartlett Learning; 2014.
39. Celentano DD, Szklo M. Gordis epidemiology. Philadelphia: Elsevier, Inc; 2018.
40. Hennekens CH, Buring JE. Epidemiology in Medicine. 1st ed. Philadelphia: Lippincott Williams & Wilkins; 1987.
41. Guo Y, Chen S-D, You J, Huang S-Y, Chen Y-L, Zhang Y, et al. Multiplex cerebrospinal fluid proteomics identifies biomarkers for diagnosis and prediction of Alzheimer's disease. Nat Hum Behav. 2024;8:2047–66.
42. Zinzuwadia AN, Mineeva O, Li C, Farukhi Z, Giulianini F, Cade B, et al. Tailoring risk prediction models to local populations. JAMA Cardiol. 2024;9:1018–28.
43. Guan C, Gong A, Zhao Y, Yin C, Geng L, Liu L, et al. Interpretable machine learning model for new-onset atrial fibrillation prediction in critically ill patients: a multi-center study. Crit Care. 2024;28:349.
44. Ganda A, Magnusson M, Yvan-Charvet L, Hedblad B, Engström G, Ai D, et al. Mild renal dysfunction and metabolites tied to low HDL cholesterol are associated with monocytosis and atherosclerosis. Circulation. 2013;127:988–96.
45. Szeto C-C, Chow K-M, Woo K-S, Chook P, Ching-Ha Kwan B, Leung C-B, et al. Carotid intima media thickness predicts cardiovascular diseases in Chinese predialysis patients with chronic kidney disease. J Am Soc Nephrol. 2007;18:1966–72.
46. Akinkuolie AO, Buring JE, Ridker PM, Mora S. A novel protein glycan biomarker and future cardiovascular disease events. J Am Heart Assoc. 2014;3:e001221.
47. SCORE2-Diabetes Working Group and the ESC Cardiovascular Risk Collaboration. SCORE2-Diabetes: 10-year cardiovascular risk estimation in type 2 diabetes in Europe. Eur Heart J. 2023;44:2544–56.
48. Ott C, Jacobs K, Haucke E, Navarrete Santos A, Grune T, Simm A. Role of advanced glycation end products in cellular signaling. Redox Biol. 2014;2:411–29.
49. Ma N, Wang H, Li Q, Chang M, Zhu J, Nan S, et al. Gut-derived IL-13 contributes to growth via promoting hepatic IGF-1 production. Microbiome. 2024;12:248.
50. Hotamisligil GS. Inflammation and metabolic disorders. Nature. 2006;444:860–7.
51. Wolf M, Böhm S, Brand M, Kreymann G. Proinflammatory cytokines interleukin 1β and tumor necrosis factor α inhibit growth hormone stimulation of insulin-like growth factor I synthesis and growth hormone receptor mRNA levels in cultured rat liver cells. Eur J Endocrinol. 1996;135:729–37.
52. Cutrín JC, Zingaro B, Camandola S, Boveris A, Pompella A, Poli G. Contribution of gamma glutamyl transpeptidase to oxidative damage of ischemic rat kidney. Kidney Int. 2000;57:526–33.
53. Aberkane H, Stoltz J-F, Galteau M-M, Wellman M. Erythrocytes as targets for gamma-glutamyltranspeptidase initiated pro-oxidant reaction. Eur J Haematol. 2002;68:262–71.
54. Milnerowicz H, Bukowski R, Jabłonowska M, Ściskalska M, Milnerowicz U. The antioxidant profiles, lysosomal and membrane enzymes activity in patients with acute pancreatitis. Mediat Inflamm. 2014;2014:376518.
55. Djuricic I, Calder PC. Pros and cons of long-chain omega-3 polyunsaturated fatty acids in cardiovascular health. Annu Rev Pharmacol Toxicol. 2023;63:383–406.
56. Yates CM, Calder PC, Ed Rainger G. Pharmacology and therapeutics of omega-3 polyunsaturated fatty acids in chronic inflammatory disease. Pharmacol Ther. 2014;141:272–82.

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